part-a

```
import pandas as pd

# Load dataset
df = pd.read_csv("data.csv")

# Display first rows
print(df.head())

# Check missing values
print(df.isnull().sum())

# Handle missing values
df = df.dropna()   # or df.fillna(method='ffill')

# Feature selection
df = df[['Age', 'Salary', 'Purchase_Amount', 'Category']]
```

```
   Age  Salary  Purchase_Amount   Category
0   56   68136             1077  Groceries
1   46   43625            41345   Clothing
2   32  102873            21611  Groceries
3   60  120843            17984  Furniture
4   25   92592            15060  Furniture
Age                0
Salary             0
Purchase_Amount    0
Category           0
dtype: int64
```

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error

# Features & target
X = df[['Age', 'Salary']]
y = df['Purchase_Amount']

# Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Model
model = LinearRegression()
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Evaluation
mse = mean_squared_error(y_test, y_pred)
print("MSE:", mse)
```

```
MSE: 199135352.7789048
```

```
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

# Encode categorical target
le = LabelEncoder()
df['Category'] = le.fit_transform(df['Category'])

# Features & target
X = df[['Age', 'Salary']]
y = df['Category']

# Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Model
clf = DecisionTreeClassifier()
clf.fit(X_train, y_train)

# Prediction
y_pred = clf.predict(X_test)

# Evaluation
acc = accuracy_score(y_test, y_pred)
print("Accuracy:", acc)
```

```
Accuracy: 0.225
```

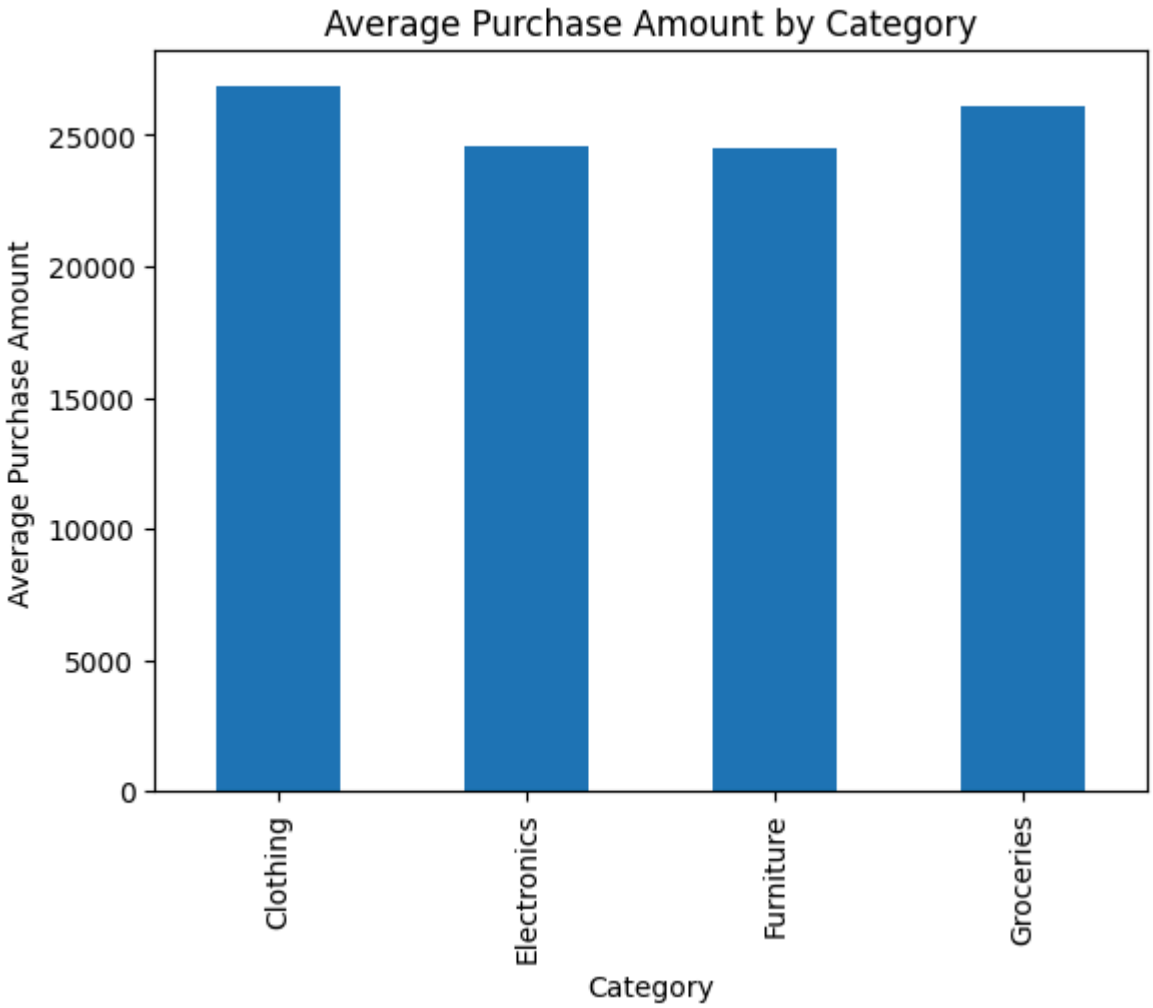
part-b

```
import pandas as pd

df = pd.read_csv("data.csv")

# Group by Category and calculate mean Purchase_Amount
grouped = df.groupby("Category")["Purchase_Amount"].mean()

print(grouped)
```

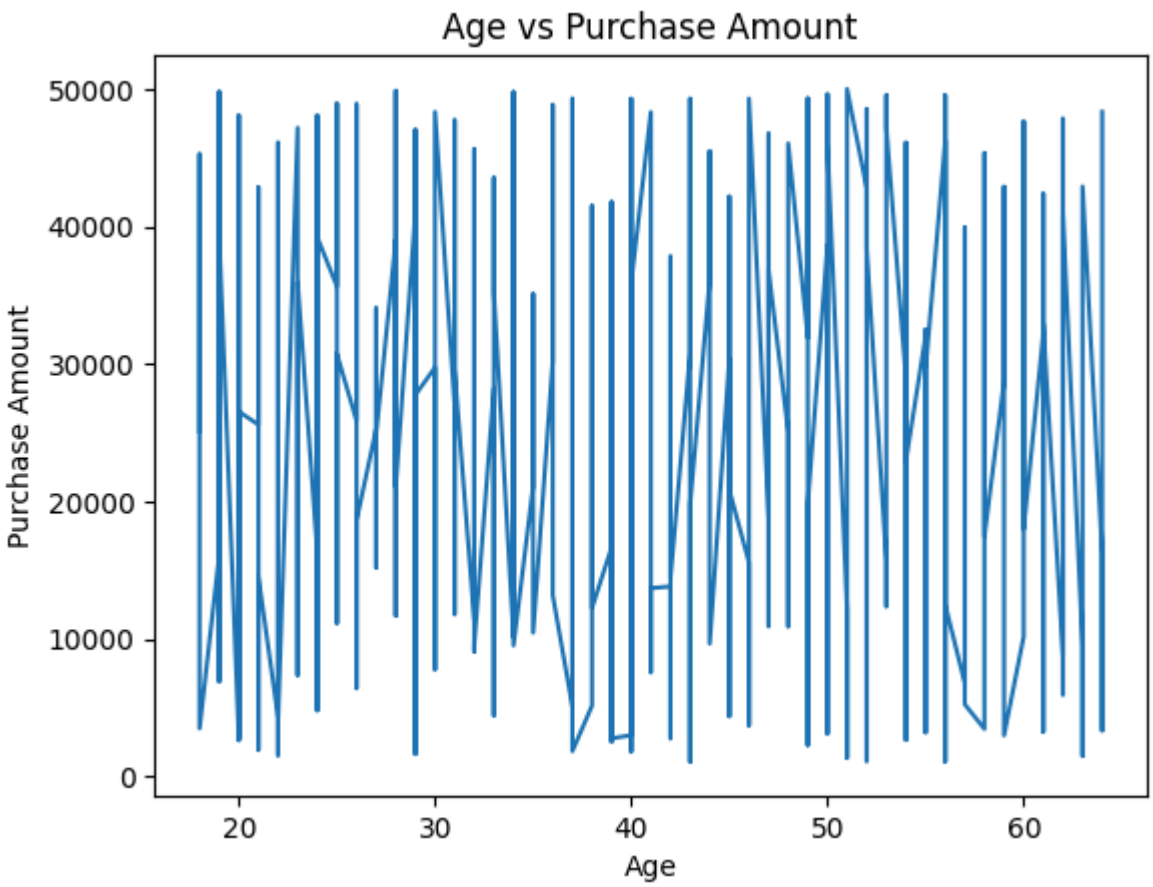
```
Category
Clothing       26857.509804
Electronics    24601.784722
Furniture      24532.759494
Groceries      26127.068966
Name: Purchase_Amount, dtype: float64
```

```
import matplotlib.pyplot as plt

grouped.plot(kind='bar')

plt.title("Average Purchase Amount by Category")
```

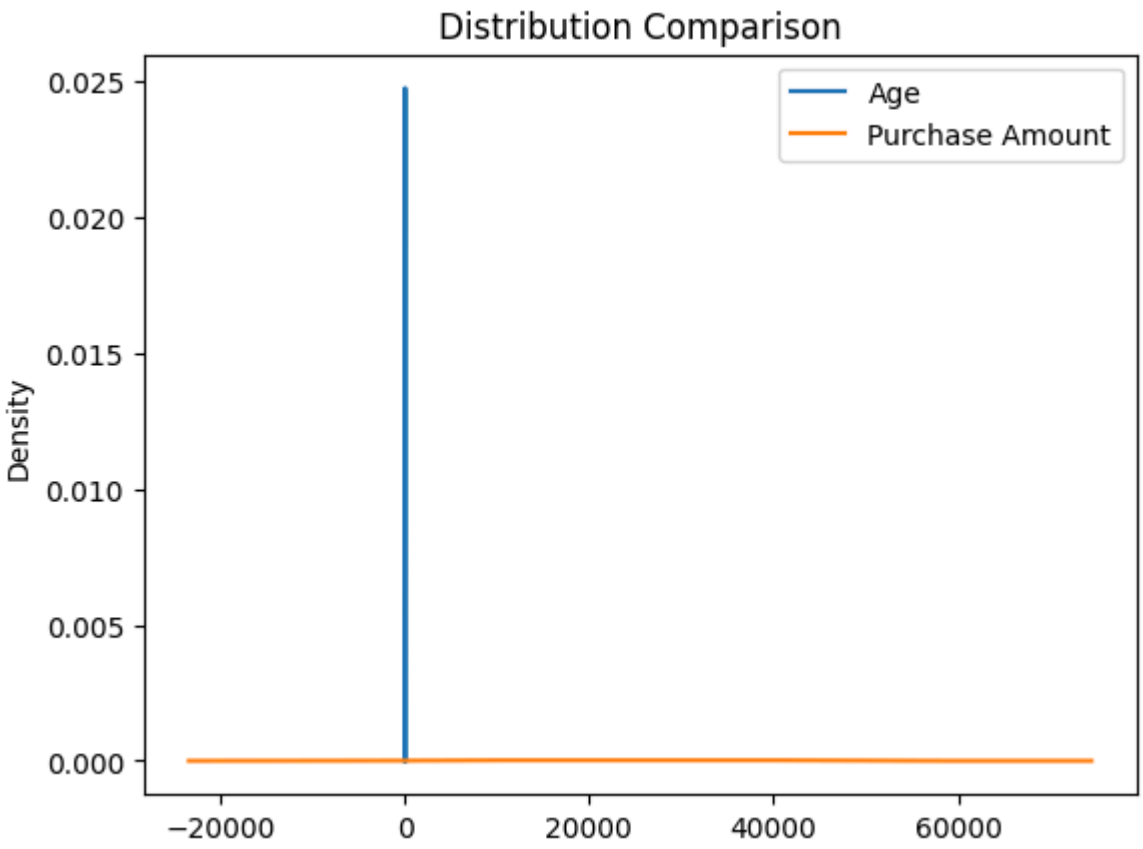
```
plt.xlabel("Category")
plt.ylabel("Average Purchase Amount")

plt.show()
```




```
df_sorted = df.sort_values(by="Age")

plt.plot(df_sorted["Age"], df_sorted["Purchase_Amount"])

plt.title("Age vs Purchase Amount")
plt.xlabel("Age")
plt.ylabel("Purchase Amount")

plt.show()
```




```
df["Age"].plot(kind='kde', label="Age")
df["Purchase_Amount"].plot(kind='kde', label="Purchase Amount")

plt.legend()
plt.title("Distribution Comparison")

plt.show()
```



part-c

```
df = pd.read_csv("data.csv")

avg = df["Salary"].mean()

filtered = df[df["Salary"] > avg]

print(filtered.head())
```

```
    Age  Salary  Purchase_Amount   Category
2    32  102873            21611  Groceries
3    60  120843            17984  Furniture
4    25   92592            15060  Furniture
5    38   97052            18449   Clothing
13   57  113105             8804  Groceries
```

```
df.describe()
```

	Age	Salary	Purchase_Amount
count	600.000000	600.000000	600.000000
mean	41.405000	83638.398333	25527.428333
std	13.230623	39398.738826	14014.327833
min	18.000000	20055.000000	1060.000000
25%	30.000000	47510.500000	13236.000000
50%	42.000000	81579.000000	25480.500000
75%	52.000000	120000.500000	37419.750000
max	64.000000	149922.000000	49988.000000

part-d

```
import pandas as pd
df = pd.read_csv("data.csv")
```

```
df.head()
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 600 entries, 0 to 599
Data columns (total 4 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Age              600 non-null    int64
 1   Salary           600 non-null    int64
 2   Purchase_Amount  600 non-null    int64
 3   Category         600 non-null    object
dtypes: int64(3), object(1)
memory usage: 18.9+ KB
```

```
df = df.dropna()
```

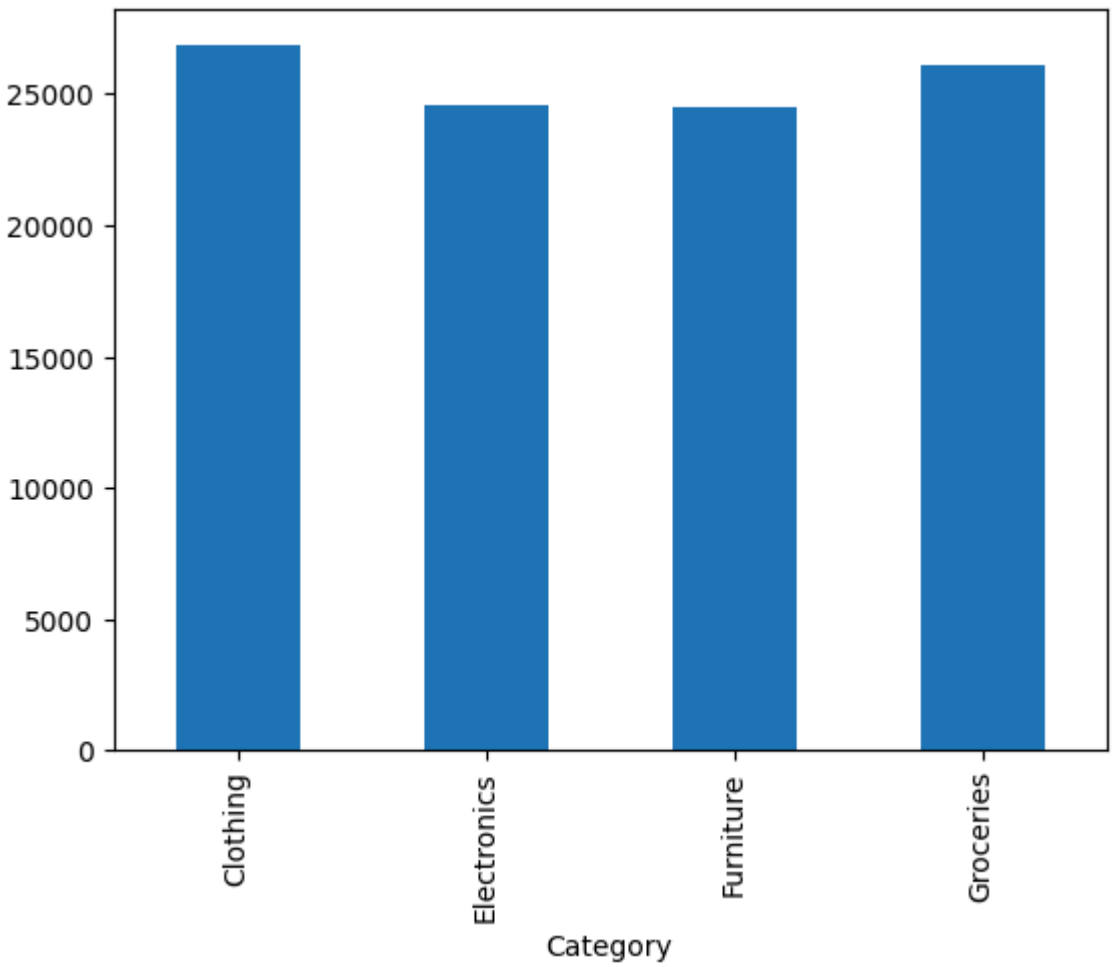
```
df.describe()
```

	Age	Salary	Purchase_Amount
count	600.000000	600.000000	600.000000
std	13.230623	39398.738826	14014.327833
min	18.000000	20055.000000	1060.000000
25%	[illegible]	[illegible]000	13236.000000
50%	42.000000	81579.000000	25480.500000
75%	[illegible]	[illegible]000	37419.750000
max	64.000000	149922.000000	49988.000000

```
df.groupby("Category")["Purchase_Amount"].mean()
```

```
Purchase_Amount
Category
Clothing        26857.509804
Electronics     24601.784722
Furniture       24532.759494
Groceries       26127.068966
dtype: float64
```

```
import matplotlib.pyplot as plt

df.groupby("Category")["Purchase_Amount"].mean().plot(kind='bar')
plt.show()
```




```
plt.plot(df["Age"], df["Purchase_Amount"])
plt.show()
```